

News

TAOH 12-2

LSC - Find
Study Partners

Introduction to Rotations

- Formalisms
- Basic Rotations
- Examples

HW2 Due Friday
6pm

MATLAB OnRamp

Formalisms

Rotation Matrix

eg.

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

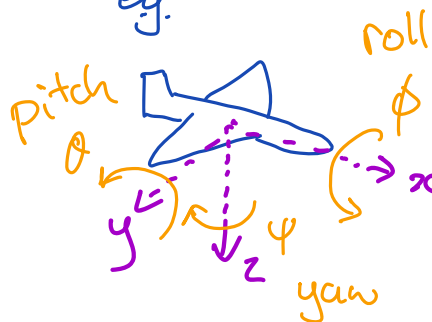
- Good Calculation
- Hard visualize
- Columns are unit vectors
" " orthogonal
∴ 3 DoFs

$$R^T = R^{-1}$$

$$R = \sum_{i=1}^3 \sum_{j=1}^3 R_{ij} \hat{e}_i \otimes \hat{e}_j$$

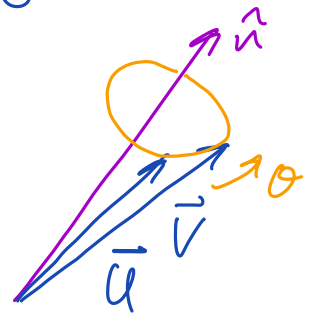
Euler Angles

eg.



- 3 rotations, 3 axes
- 12 possible combinations
 - 6 'Proper' $xyx, xzx, yxy, yzy, zxz, zyz$.
 - 6 'Tait-Bryan' $xyz, xzy, yxz, yzx, zxy, zyx$.
- Can define $R_x R_y R_z = R$

Angle-Axis



- 1 rotation, 1 axis
- Used in Rodrigues Rotation Form.
- $(\hat{n}, \theta)$ found from $\text{eig}(R)$ and $\text{Trace}(R)$

Quaternions

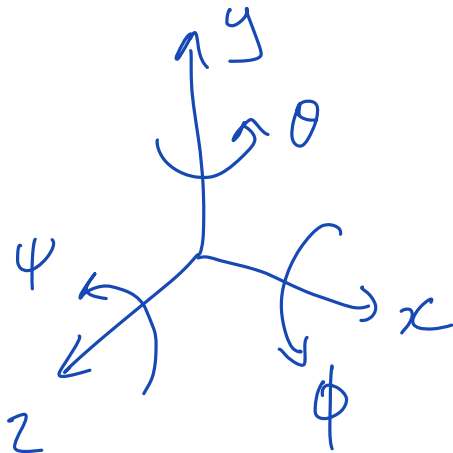
$$q = q_r + q_1 i + q_2 j + q_3 k$$

 $\Rightarrow$ smooth interpolation

3 Blue 1 Brown
MAE 6700

The Mad Hatter's Tea Party

Basic Rotation



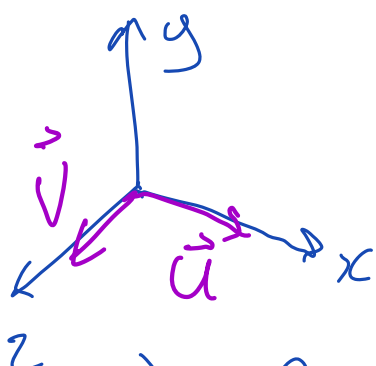
$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}$$

Direction
Cosine
Matrices
(DCMs)

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_z(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Ex 1 Find two R that rotate $\hat{u}$ to $\hat{v}$



i) directly (R_1)

$$V = RU$$

ii) through 1 intermediate frame (R_2)

i) $\theta = -90^\circ$

$$R_1 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

ii)

$$R_x(90^\circ) R_z(90^\circ)$$

$$R_2 = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

Ex 2 R_1 does not truly equal R_2 , as it neglects orientation. Instead use 2 intermediate frames in order to find a true equal.

Instructor Ans: $R_3 = R_x(90^\circ) R_y(-90^\circ) R_z(90^\circ) = R_1$

Student Ans: $R_4 = R_z(-90^\circ) \underbrace{R_x(90^\circ) R_z(90^\circ)}_{R_2} = R_1$

↑ Thanks!

Note: You can use any number of rotations ($R_1 R_2 R_3 \dots R_n = R_A$) to find a rotation that matches R_1 , however to truly match it must hold true for any vector

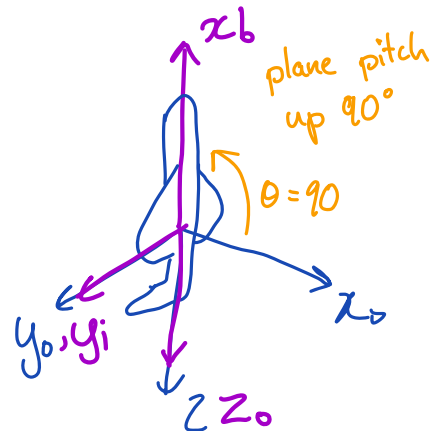
i.e. $R_1 = R_A$ if and only if $(R_1 \vec{a} = R_A \vec{a}) \& (R_1 \vec{b} = R_A \vec{b}) \& (\vec{b} \neq \vec{a})$

These are examples of extrinsic Euler angles, where rotation is about a fixed coordinate system.

$\begin{cases} R_3: zyx \text{ - Tait-Bryan} \\ R_4: zxz \text{ - proper} \end{cases}$

Aircraft typically use intrinsic Euler angles, where the rotation axes move with the body

Intrinsic Euler angles are easily measured by instrument, but are prone to gimbal lock.



Exercise 3

Find a single axis that rotates the cube to each of the 4 positions below. Draw them on the diagram

[Bonus, write down the unit vector $\hat{n}$ and rotation angle θ]

Starting point.

